# Synchronous phasors and the power angle

Answers and worked solutions · Day 14 · Fri 6 Nov · pack M11

## 5. Practice

### From your course files

**PQ 8.8 (practice questions).**

a. $V = 1\angle 0°$. Rated VA at unity pf gives $I = 1\angle 0°$ pu.

$$E_0 = 1 + j1.8(1\angle 0°) = 1 + j1.8 = 2.059\angle 60.95° \text{ pu}$$

**Excitation = 2.059 pu.**

b. **Rotor angle $\delta = 60.95°$.**

c. $P_{max} = \frac{VE_0}{X_S} = \frac{1\times 2.059}{1.8}$ = **1.144 pu** at $\delta = 90°$.

All three match the course answers.

**Jun24 Q4b (past exam, 6 marks).**

i. [3] $V = 1\angle 0°$ pu. Rated VA means S = 1 pu, so $|I| = 1$ pu. Leading: $\phi = +\cos^{-1} 0.9 = +25.84°$. So $I = 1\angle +25.84°$ pu.

$$E_0 = V + jX_S I = 1\angle 0° + (1.8\angle 90°)(1\angle 25.84°) = 1 + 1.8\angle 115.84° = 0.2155 + j1.620$$

$$E_0 = \mathbf{1.634\angle 82.42°} \text{ pu}$$

ii. [1] **Rotor angle $\delta = 82.42°$.** This is very close to the 90° limit, because the machine is under-excited.

iii. [2] With $E_0$ and V fixed, $P_{max} = \frac{VE_0}{X_S} = \frac{1\times 1.634}{1.8}$ = **0.9077 pu**, at a **rotor angle of 90°**.

(See Figure 1.) Unrounded working gives 82.43° and 0.9079 pu. The course's 82.42° and 0.9077 pu come from rounding $E_0$ to 1.634 first. Either is fine.

**PQ 8.9 (practice questions).**

Given $E_0 = 1.5$ pu, $V = 1.02$ pu, $X_S = 1.8$ pu, $\delta = 70°$:

$$P = \frac{VE_0}{X_S}\sin\delta = \frac{1.02 \times 1.5}{1.8}\sin 70° = 0.7987 \text{ pu}$$

$P = 0.7987 \times 200 =$ **159.7 MW**. This matches the course answer.

**PQ 8.11 (practice questions).**

a. $Z_B = \frac{15^2}{250} = 0.9\ \Omega$. $|Z_S| = \frac{5750/\sqrt{3}}{2050} = 1.619\ \Omega$.
$X_S = \sqrt{1.619^2 - 0.15^2} = 1.612\ \Omega$, so $X_S = 1.791$ **pu**. $R = 0.15/0.9 = 0.1667$ pu.

b. Full load means S = 1 pu at V = 1 pu, so $|I| = 1$ pu. 0.85 lag: $I = 1\angle -31.79^\circ$ **pu**.

c. $E_0 = 1 + (1\angle -31.79^\circ)(0.1667 + j1.791) = 1 + 1.799\angle 52.89^\circ = 2.531\angle 34.53^\circ$ **pu**.

d. $E_{0,line} = 2.531 \times 15 =$ **37.97 kV**.

e. On the air-gap line, 50 A gives 5.75 kV, so $I_F = 37.97\ \text{kV} \times \frac{50}{5.75\ \text{kV}} =$ **330.2 A**.

f. $\delta = 34.53^\circ$. All six match the course answers.

**PQ 8.12 (practice questions).**

a. $Z_B = \frac{16^2}{120} = 2.133\ \Omega$. $|Z_S| = \frac{4400/\sqrt{3}}{600} = 4.234\ \Omega$.
$X_S = \sqrt{4.234^2 - 0.1^2} = 4.233\ \Omega$, so $X_S = 1.984$ **pu**. $R = 0.1/2.133 = 0.0469$ pu.

b. $P = 102/120 = 0.85$ pu. $S = 0.85/0.95 = 0.8947$ pu. At V = 1 pu:
$I = 0.8947\angle -18.19^\circ$ **pu**.

c. $E_0 = 1 + (0.8947\angle -18.19^\circ)(0.0469 + j1.984) = 2.311\angle 46.39^\circ$ pu.

The synchronising field current (OC voltage = 16 kV) is $I_F = 50 \times \frac{16}{4.4} = 181.8$ A.

So $I_F = 2.311 \times 181.8 =$ **420.2 A**, and the **ratio is 2.31 : 1** ($= |E_0|$ in pu).

The course gives 420.4 A and $2.312\angle 46.40^\circ$. The 0.2 A difference is only rounding: with unrounded values I get $E_0 = 2.3112$ pu, and $2.3112 \times 181.8 = 420.2$ A. Either is acceptable.

**PQ 8.13 (practice questions).**

$P_E = 270/0.92 = 293.5$ kW.

$|Q|_{max} = \sqrt{330^2 - 293.5^2} =$ **150.9 kVAr**. It can be either sign: −150.9 kVAr over-excited (supplying Q for PFC) or +150.9 kVAr under-excited. This matches the course answer.

**PQ 8.14 (practice questions).**

This is worked in full as Example 3 (with the plant figure). The key lines:

1. Induction motor: $P = 200/0.9 = 222.2$ kW, $S = 222.2/0.82 = 271.0$ kVA, $Q = +155.1$ kVAr.
2. Synchronous motor: $P = 80$ kW, $Q = -\sqrt{100^2 - 80^2} = -60$ kVAr (over-excited).
3. Totals: $P = 302.2$ kW, $Q = +95.1$ kVAr, so $\text{pf} = \cos(\tan^{-1}(95.1/302.2)) =$ **0.9539 lag**.

This matches the course answer.

**Jun23 Q4b (past exam, 7 marks).**

i. [2] Remaining induction motor: $P_{IM} = 200/0.85 =$ **235.3 kW**. $S = 235.3/0.83 = 283.5$ kVA. $Q_{IM} = 283.5\sin(\cos^{-1}0.83) =$ **+158.1 kVAr** (lagging).

ii. [2] Synchronous motor: $P_{SM} = 200/0.90 =$ **222.2 kW**. The most leading Q within the 250 kVA rating is $Q_{SM} = -\sqrt{250^2 - 222.2^2} = -114.5$ kVAr. So the **capability is 114.6 kVAr** (supplied, leading).

iii. [3] $P_{tot} = 235.3 + 222.2 = 457.5$ kW. $Q_{tot} = 158.1 - 114.5 = 43.6$ kVAr.

$$\text{pf} = \cos\left(\tan^{-1}\frac{43.6}{457.5}\right) = \mathbf{0.9955\ lag}$$

All the parts match the answer sheet. The sheet's 114.6 kVAr is a rounding of 114.53.

**PQ 8.15 (practice questions).**

- Existing load: $P = 2760 \times 0.71 = 1960$ kW, $Q = 2760\sin(\cos^{-1}0.71) = +1944$ kVAr.
- Induction motors (800 kW mechanical): $P = 800/0.89 = 898.9$ kW, $Q = 898.9\tan(\cos^{-1}0.75) = +792.7$ kVAr.
- Synchronous motors (the remaining 700 kW mechanical): $P = 700/0.93 = 752.7$ kW, $Q = -\sqrt{1350^2 - 752.7^2} = -1121$ kVAr.
- Totals: $P = 3611$ kW (course: 3612 kW), $Q = 1616$ kVAr.

$\text{pf} = \cos(\tan^{-1}(1616/3611)) =$ **0.9128 lag**. The course says 0.9129. That is the last-digit rounding of its rounded totals.

**PQ 8.10 (practice questions).**

a. $I_B = \frac{150\text{ M}}{\sqrt{3}\times 15\text{ k}} =$ **5774 A**. $Z_B = 15^2/150 =$ **1.50 Ω**.

b. At $I_F = 1$ A: $E_{ph} = 500/\sqrt{3} = 288.7$ V and $I_{SC} = 100$ A, so $|Z_S| = 2.887\ \Omega$. $X_S = \sqrt{2.887^2 - 0.3^2} = 2.871\ \Omega$ = **1.914 pu**.

c. Synchronising needs $E_0 = 15$ kV line, so $I_F = 15000/500 =$ **30 A**.

d. **1 pu** (by definition).

e. $I_F = 1.65 \times 30 =$ **49.5 A**. All match the course answers.

**Jun23 Q4c (past exam, 8 marks).**

These are built from the answer sheets' points. The sheet says you do not need every point for full marks. Aim for about one solid point per mark.

**(i) Power-factor correction [3]**

- Many loads, such as induction motors, draw reactive power as well as real power. Their current lags the voltage.
- Reactive power is energy oscillating back and forth between the source and the load. It does no useful work, but it adds to the current.
- The extra current loads up generators, transformers and cables, causes extra resistive losses and voltage drop, and raises the VA demand that the supply company charges for.
- Correction means supplying the reactive power locally. Capacitors, or an over-excited synchronous motor, are connected in parallel with the load and generate reactive power. The supply then provides mostly real power, and the pf moves towards unity.
- The useful result: lower current and losses for the same useful power, freed-up capacity, and lower tariffs.

**(ii) Transformers vs induction machines [3]**

- Similarities: both can run from three-phase AC, have primary (stator) and secondary (rotor) windings on a magnetic core, and transfer power by mutual induction. Their equivalent circuits have the same form: winding resistances, leakage reactances, a magnetising reactance and a core-loss resistance.
- Differences: the induction machine's secondary is a rotating, short-circuited winding. Its rotor frequency is the slip frequency, not the stator frequency. It has an air gap, so its magnetising current is much larger.
- In the induction-machine equivalent circuit, the referred rotor resistance is divided by the slip. That term represents the mechanical output plus the rotor loss.

**(iii) Perfectly conducting rotor cage [2]**

- The referred rotor resistance would be zero.

- The torque–speed curve would be vertical at synchronous speed, like a synchronous machine. The machine could only produce torque at synchronous speed.
- There would be no rotor copper losses, so the efficiency would be higher.
- The starting torque would be zero, so the motor could not start by itself.

**Jun24 Q4c (past exam, 9 marks).**

These are built from the answer sheets' points. The sheet says you do not need every point for full marks. Aim for about one solid point per mark.

**(i) Real, reactive and apparent power [3]**

- Real power: produced by the part of the current that is in phase with the voltage. It is the average of the instantaneous power, it is energy that flows one way from source to load, and it does useful work. Resistors absorb it.
- Reactive power: produced by the part of the current that is 90° out of phase with the voltage. Its average is zero, because energy oscillates between source and load each cycle, so it does no useful work. Inductors absorb it and capacitors generate it. It still adds to the current and the losses.
- Both real and reactive power are scalar quantities.
- Apparent power: the magnitude of the complex power, whose real part is the real power and whose imaginary part is the reactive power. It is the product of rms voltage and total current, so it sets the current and the ratings (VA) of generators, transformers and cables.

**(ii) Transformer losses [3]**

- Copper (winding) loss: resistive heating in the primary and secondary windings. It is modelled by the series winding resistances. It is proportional to current squared, so it rises with the square of the output power.
- Iron (core) loss: hysteresis and eddy-current loss in the core, caused by the alternating flux. It is modelled by the parallel core-loss resistance. It depends on the voltage and frequency, not the load, so it is roughly constant with output power.

**(iii) Why induction-machine no-load current is not zero [3]**

- Magnetising current is needed to set up the rotating air-gap flux. Because there is an air gap, this current is large. It is modelled by the magnetising reactance.
- Core-loss current supplies the hysteresis and eddy-current losses in the stator and rotor iron. It is modelled by the core-loss resistance.
- Even at no load, the rotor needs some torque to overcome mechanical losses: bearing friction, and windage (air drag and the cooling fan). So it runs slightly below synchronous speed with a small rotor current. That draws a little real power.
- So the no-load current is mostly reactive (a low pf), plus a small real component.

**Practice Questions, Ch8 short answers (j) and (k) (bonus).**

- j. Reactive power capability is limited mainly by the stator VA rating, because of stator winding heating. At a real power P, the most Q available is the square root of (rated S squared minus P squared). At full rated real power, no reactive power is left. When absorbing Q (under-excited), the machine is also limited by the rotor angle approaching the stability limit.
- k. Industrial plant pf is usually lagging, because most loads are induction motors that need magnetising current. Synchronous motors are used because, when over-excited, they draw leading current and supply reactive power. That corrects the plant pf and lowers the VA charges, while also driving a constant-speed load.

## New questions

**N1.**

With P = 0, the current is at ±90° to V and $\delta = 0$, so everything is in line.

a. Supplying Q from the machine means a lagging output current: $I = 0.8\angle -90^\circ = -j0.8$ pu.

$E_0 = 1 + j1.5(-j0.8) = 1 + 1.2 =$ **2.2 pu (over-excited)**.

Check with the derived formula: $Q = (1 \times 2.2 - 1)/1.5 = 0.8$ pu.

b. Absorbing Q means a leading output current: $I = +j0.5$, so $E_0 = 1 + j1.5(j0.5) = 1 - 0.75 =$ **0.25 pu (under-excited)**.

**N2.**

a. Per phase: $P_{ph} = \frac{(11000/\sqrt{3})(14000/\sqrt{3})}{2.4}\sin 25^\circ = 9.039$ MW. So **P = 3 × 9.039 = 27.12 MW**.

With line quantities: $P = \frac{11000 \times 14000}{2.4}\sin 25^\circ = 27.12$ MW, the same.

b. $P_{max} = \frac{11000 \times 14000}{2.4} =$ **64.17 MW** at $\delta = 90^\circ$.

In pu: $Z_B = 11^2/50 = 2.42\ \Omega$, so $X_S = 0.9917$ pu and $E_0 = 14/11 = 1.273$ pu. Then $P_{max} = \frac{1 \times 1.273}{0.9917} =$ **1.283 pu** (= 64.17/50). The pu formula already gives three-phase power.

**N3.**

a. $I = 0.8\angle -31.79^\circ$ pu. $E_0 = 1 + j1.6(0.8\angle -31.79^\circ) = 1 + 1.28\angle 58.21^\circ = 1.997\angle 33.02^\circ$ **pu**.

b. Same P = 0.8 × 0.85 = 0.68 pu, so $|I|$ stays 0.8 pu, but now $I = 0.8\angle +31.79^\circ$.

$E_0 = 1 + 1.28\angle 121.79^\circ = 1.136\angle 73.33^\circ$ **pu**.

c. Lagging: $P_{max} = 1.997/1.6 =$ **1.248 pu**. Leading: $P_{max} = 1.136/1.6 =$ **0.710 pu**.

Check: $\frac{1.136}{1.6}\sin 73.33^\circ = 0.68$ pu, as required.

Comment: reducing the excitation (to absorb Q) cuts $P_{max}$ from 1.25 to 0.71 pu, and pushes $\delta$ from 33° to 73°. That is right at the typical 75° operating limit. The machine has only about 4% power margin before pull-out, compared with about 84% when lagging.

**N4.**

a. $E_{0,min} = \frac{PX_S}{V\sin 75^\circ} = \frac{0.6\times 1.8}{\sin 75^\circ} =$ **1.118 pu**.

b. $I = \frac{E_0 - V}{jX_S} = \frac{1.118\angle 75^\circ - 1}{j1.8} = 0.718\angle +33.34^\circ$ **pu** (leading).

$S = 0.718$ pu. $Q = S\sin(-\phi) = 0.718\sin(-33.34^\circ) =$ **−0.395 pu**, so the machine absorbs 0.395 pu.

Check with the derived formula: $Q = \frac{1.118\cos 75^\circ - 1}{1.8} = -0.395$ pu.

pf = 0.6/0.718 = **0.835 leading**.

**N5.**

a. $P_E = 300/0.95 =$ **315.8 kW**. $Q_{SM} = -\sqrt{400^2 - 315.8^2} =$ **−245.5 kVAr** (supplied).

b. Plant: $Q_p = 600\tan(\cos^{-1}0.75) = +529.2$ kVAr.

$P_{tot} = 915.8$ kW, $Q_{tot} = 529.2 - 245.5 = 283.6$ kVAr.

pf = $\cos(\tan^{-1}(283.6/915.8)) =$ **0.9552 lag** (it was 0.75 before).

N5(c) motor circuit: input current 1.000 at +37.86 degrees through j1.0 pu

c. $S_m = (315.8 - j245.5)/400 = 0.7895 - j0.6138$ pu, with $|S_m| = 1$ pu. At V = 1 pu, $I_m = (S_m/V)^* = 1.000\angle +37.86^\circ$ **pu** (leading). That is the rated current, $\frac{400\text{ k}}{\sqrt{3}\times 415} =$ **556.5 A**.

$E_0 = V - jX_S I_m = 1 - (1\angle 90^\circ)(1\angle 37.86^\circ) = 1.797\angle -26.07^\circ$ **pu**, which is $1.797 \times 415 =$ **745.6 V (line)**. δ is negative, so it is motoring.

d. Unity pf: $I_m = 0.7895\angle 0^\circ$, so $E_0 = 1 - j0.7895 = 1.274\angle -38.29^\circ$ **pu**.

The motor now contributes Q = 0, so plant pf = $\cos(\tan^{-1}(529.2/915.8)) =$ **0.866 lag**. Lower excitation means less correction and also a larger $|\delta|$.

## 6. Quick recall check

**Question**

1. At $\delta = 90^\circ$ (the theoretical stability limit). Past it, the machine loses synchronism (pull-out), with large currents and oscillating torques. In practice it is run at about 75° or less.
2. Under-excited. It absorbs Q.
3. Three-phase. Multiply by $S_{base}$ to get watts.
4. $\sqrt{100^2 - 60^2} = 80$ kVAr, which is −80 kVAr in the course sign convention.
5. At the bottom of the V. The pf there is unity (Q = 0), so $|I| = P/V$.